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Student Name: _____ Student ID: _____

THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Course : Broad Discipline of COMP-61431

Subject : COMP3423 Human Computer Interaction

Group : 1011, 1012

Session : 2022 / 2023 Semester I

Date : 12 December 2022 Time : 12:30 - 14:30

Time Allowed: 2 Hours Subject Lecturer: Prof. HOORN Johan F/ Dr TANG Jeff

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This question paper has ___18___ pages (cover included).
(Some pages may be intentionally omitted.)

Instructions to Candidates:

- This exam has TEN multiple choice items and FIVE open questions.
- Attempt ALL questions.
- Write your answers in a reasonable neat and coherent way, in the space provided.
- In case you run out of space, you may use the back of the page; in this case please indicate clearly.
- Write down your name and student numbers on each page on this question/answer book.
- You are not allowed to bring books, articles, print-outs, or electronic equipment.

Do not turn this page until you are told to do so!

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Section B: Open Questions [80 marks]

Answer all FIVE questions in this section.

Question 11: HCI Knowledges [10 marks]

(a) (3 marks) How does desktop virtual reality differ from immersive virtual reality?

(b) (4 marks) Give examples of **two** questions that a participant may ask during a thinking-aloud evaluation.

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(c) (3 marks) What are usability heuristics?

Question 12: Human, Haptics and Touch [20 marks]

(a) (4 marks) What is haptic human-computer interaction? Name **one** technology and describe how it enables haptics in HCI.

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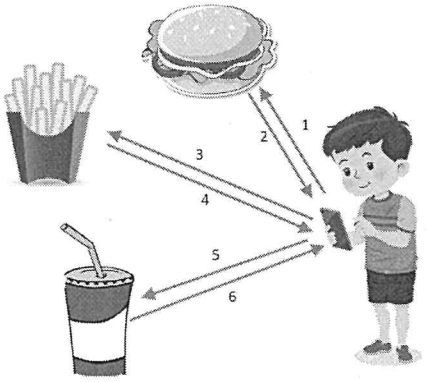
(b) (5 marks) Within the rubber-hand experiment setup, describe how "body transfer illusion" and "sense of embodiment" are induced.

(c) FoodTiger is a food delivery app. One day, James purchases a hamburger meal for lunch – *a hamburger meal contains a hamburger, a pack of french fries, and a cup of soft drink*. The following figure shows how James selects the food items (the arrows shows the sequence of selection) is required to tap "burgers", and then all burgers are shown on a popup page, for James to select. After selection, the UI goes back to the menu and James is required to select the size of french fries and drinks on different popup pages.



i. (3 marks) Name the strategy provided by FoodTiger app, for a user to buy a meal composed of multiple items of different choices? Give **one** reason to justify your answer.

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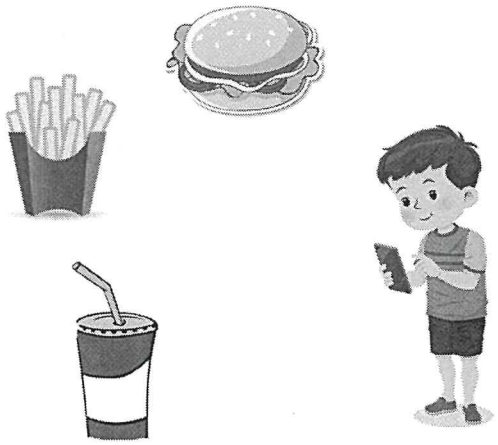
ii. (2 marks) Propose one problem-solving strategy that provides the user with an alternative way to select food items. Illustrate your strategy with arrows to show the sequence of food selection in the following figure.

Strategy name:

Sequence:



iii. (6 marks) Describe how the app would work using the problem-solving strategy you proposed in c(ii) (You may use drawings to illustrate). Discuss **one** advantage of your proposed strategy.

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Q12(c)iii. continue…

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Question 13. Dialog design [15 marks]

A digital toy camera designed for children has one physical *power switch* and two physical buttons: *filter button* and *shutter button*. When the power switch slides to on, the user can press the filter button to cycle through three filter states: (1) *No filter*, (2) *Monochromatic*, and (3) *Starry*. At any filter state, the user can press the shutter button to take a picture; afterwards the system will automatically go back to the current filter state. Anytime when the camera is on (i.e. at any filter state), the user can turn off the camera by sliding the power switch to "off". Draw a State Transition Network (STN) to describe the operations of the above camera.

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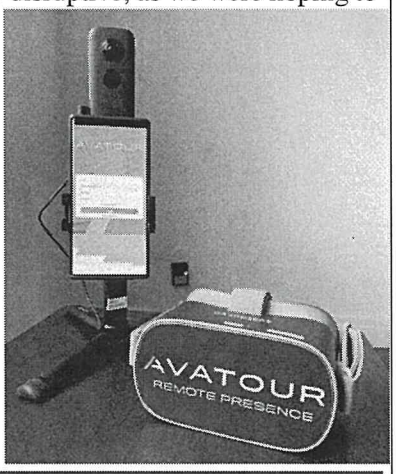
Question 14. Visual, Perception [20 marks]
Read this article and answer the following questions.

VR Platform Offers Solution to COVID-19 Travel Restrictions

A new communications technology is helping tech-forward companies respond to COVID-19 travel restrictions. The AVATOUR remote presence platform, created by San Francisco startup Imeve, employs the latest virtual reality technology to transport users to a remote location in real-time, providing a new and effective substitute for travel. Shenzhen-based camera maker Insta360 builds the 360-degree. cameras that make AVATOUR possible. Together, the two companies are providing a new alternative for business people who find themselves suddenly unable to visit facilities worldwide.

AVATOUR is a cloud service that enables real-time, multi-user, 360-degree. remote presence. Ken Minn, CEO of smartphone accessory company Kenu, was one of the first to use AVATOUR to virtually visit a manufacturing partner in China. "Before the coronavirus, someone from our team would be in China at least once a month," said Minn. "The outbreak has been pretty disruptive, as we were hoping to ship a new product immediately after Lunar New Year. With AVATOUR, we are able to walk through the entire factory with our Chinese partners to review progress, just as if we're there. In some ways it's even better than before—now I can visit anytime without the jet lag."



AVATOUR integrates with Insta360's ONE X camera, enabling realistic, live, 4K 360-degree. capture with an intuitive, palm-sized device. AVATOUR has been in development for two years, and in beta since Fall 2019; this new joint offering with Insta360 marks the first commercial release of the platform.

"As more companies and countries limit or halt travel, businesses worldwide are being forced to find remote working solutions. Videoconferencing platforms like Zoom and Skype can replace a conference room, but until now, there was no substitute for location-dependent meetings like facility walkthroughs, site inspections, and site-specific training," said Imeve CEO Devon Copley. "Using AVATOUR to visit a location alongside someone thousands of miles away feels more like an in-person visit than a videoconference."

The AVATOUR platform is available now to qualified companies in the United States, EU, UK, Australia, New Zealand, Japan, and China. After qualification, Insta360 will ship capture hardware directly to the facility, which will host AVATOUR sessions. Imeve provides remote visitors with VR headsets from Oculus/Facebook.

Reference: Imeve (March 12, 2020). VR Platform Offers Solution to COVID-19 Travel Restrictions, url: https://www.newequipment.com/research-and-development/article/21126021/vr-platform-offers-solution-to-covid19-travel-restrictions

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a) (i) (2 marks) Define Place Illusion (PI)
(ii) (2 marks) Define Plausibility Illusion (Psi)
(iii) (6 marks) With the information from the article, discuss how VR devices are used to create Place Illusion (PI) and Plausibility Illusion (Psi) in the given scenario.

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b) (i) (2 marks) What is motion parallax?
(ii) (4 marks) How do human eyes perceive depth? Use a drawing to illustrate.
(iii) (4 marks) How Motion Parallax used to produce 3D vision in immersive virtual reality? Use a drawing to illustrate.

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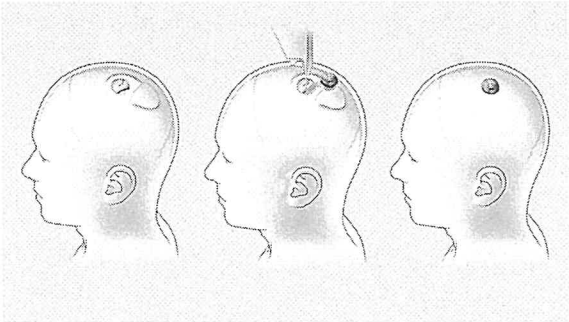
Question 15. Ergonomics and Testing [15 marks]

This question is about haptics and advanced I/O

(a) (6 marks) Peter is a university professor who prepares assignments and lecture notes in front of his desk and mainly uses mouse and keyboards over 10 hours every day. Describe any **three** related risks and how each could be improved by Ergonomics.

(b) (4 marks) The figure[1] below shows the concept of *Neuralink* that aims to develop ultra-high bandwidth brain-machine interfaces (BMI) to connect humans and computers.



What is the major risk to the human being? What is the main reason that this device is still not approved to be installed on the human body?

[1] Screenshot by Stephen Shankland/CNET url: https://www.cnet.com/science/elon-musk-shows-neuralink-brain-implant-working-in-a-pig/

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Q15(b) continue…

(c) In a usability test, a group of 40 participants were asked to evaluate two different ergo-keyboard designs.

i. (2 marks) The researchers decide to use "within-subjects design". What does this mean?

ii. (1 mark) What is a potential problem with an "Within-subjects design" setup in a usability test?

iii. (2 marks) Suggest a method to prevent the problem described in (c)ii, and describe how the above experiment could be modified.

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Q15(c) continue…

** END OF EXAMINATION **